Sex-Specific Risk Factor Analysis and Probability Estimation for Stroke

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Research Questions

- **What clinical risk factors are associated with stroke?**
  - Coronary Heart Disease (CHD)
  - Blood Pressure Medication Use
  - Smoking Status
  - Total Cholesterol
  - Body Mass Index (BMI)
- **What is the 10-year probability of stroke for specific risk profiles?**
  - Age (40, 50, 60) and Sex
  - Baseline, High Blood Pressure, Diabetes, High BP + Diabetes, Smokers
- **How do covariates change with time?**
  - Blood pressure and diabetes across 3 clinical exams

Data Overview

- 4434 Total Subjects
  - 32 excluded (prevalent stroke)
- 111 Strokes During Follow-up
  - 2.5% of all subjects
- 9.5 Years of Average Follow-up
- Minimal Missingness
  - 4275 Final Sample Size (127 w/ Missing)
- Smoking more common among males
- Covariates
  - Age
  - Systolic Blood Pressure
  - Total Cholesterol
  - BMI
  - Diabetes
  - Coronary Heart Disease
  - Blood Pressure Medication Use
  - Smoking Status

Table 1: Descriptive Statistics of Events, Follow-up, and Baseline Risk Factors

Characteristic	**Male** N = 1,930	**Female** N = 2,472	**Overall** N = 4,402
Stroke During Follow-up, n (%)	49 (2.5%)	62 (2.5%)	111 (2.5%)
Follow-up Time (Years), mean (SD)	9.40 (1.82)	9.60 (1.54)	9.51 (1.67)
Age (Years), mean (SD)	50 (9)	50 (9)	50 (9)
Systolic Blood Pressure (mmHg), mean (SD)	132 (19)	134 (24)	133 (22)
Total Cholesterol (mg/dL), mean (SD)	234 (42)	240 (46)	237 (45)
Missing, n (%)	7 (0.4%)	45 (1.8%)	52 (1.2%)
BMI (kg/m^2), mean (SD)	26.2 (3.4)	25.6 (4.5)	25.8 (4.1)
Missing, n (%)	4 (0.2%)	13 (0.5%)	17 (0.4%)
Diabetes, n (%)	59 (3.1%)	60 (2.4%)	119 (2.7%)
Coronary Heart Disease, n (%)	120 (6.2%)	67 (2.7%)	187 (4.2%)
Blood Pressure Mediation, n (%)	40 (2.1%)	96 (3.9%)	136 (3.1%)
Missing, n (%)	22 (1.1%)	38 (1.5%)	60 (1.4%)
Current Smoker, n (%)	1,168 (61%)	1,001 (40%)	2,169 (49%)

Analysis Methods

- Base Model
  - Cox Proportional Hazards Models for Males and Females
    - Risk of stroke by age, diabetes status, and systolic blood pressure
- Identify Potential Risk Factors
  - CHD, blood pressure medication, smoking status, total cholesterol, BMI
  - Selected using AIC-based backwards selection
    - Union for males and females
- Estimate 10-year probability of stroke for specific profiles
  - Male and Female
  - Ages: 40, 50, 60
  - High Blood Pressure (BP=160), Diabetes, High BP + Diabetes, Smokers
- Assess Proportional Hazards
  - Visualization and statistical testing of Schoenfeld residuals
- Descriptively Evaluate Impact of Time-Varying Covariates
  - Diabetes Status and Blood Pressure

Results

- Potential association between stroke and BP Meds/Prev CHD
- Smoking Status selected through AIC-based backward selection
  - Only significant for males
  - 92% (3%, 357%) higher hazard of stroke for males who smoke

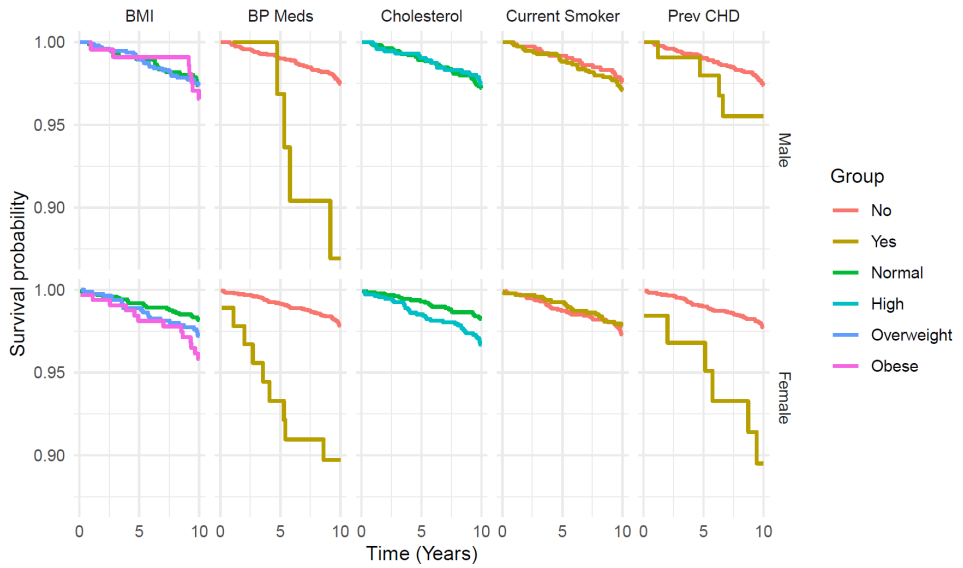

Figure 1: Kaplain-Meier Curves by Sex for Candidate Risk Factors

Table 2: Cox Proportional Hazards Model Estimates

	Male			Female		
Coefficient	Hazard Ratio	95% CI	p-value	Hazard Ratio	95% CI	p-value
Age	1.066	(1.027, 1.107)	<0.001	1.078	(1.037, 1.121)	<0.001
Diabetes	4.807	(2.099, 11.005)	<0.001	1.811	(0.652, 5.031)	0.254
Systolic Blood Pressure	1.034	(1.022, 1.046)	<0.001	1.026	(1.018, 1.035)	<0.001
Smoking Status	1.924	(1.036, 3.573)	0.038	1.435	(0.819, 2.516)	0.207

Results

Table 3: 10-Year Probability of Stroke for Males

Risk Profile	Age (Years) 40	50	60
Baseline	0.6 (0.1, 1.0)	1.1 (0.4, 1.7)	2.0 (0.8, 3.2)
Diabetes	2.7 (0.0, 5.4)	5.0 (0.7, 9.1)	9.3 (1.7, 16.3)
High BP	1.4 (0.2, 2.6)	2.7 (1.1, 4.3)	5.1 (2.4, 7.7)
High BP + Diabetes	6.7 (0.0, 13.5)	12.4 (1.8, 21.9)	22.3 (5.4, 36.2)
Smoker	1.1 (0.4, 1.7)	2.0 (1.2, 2.9)	3.8 (2.0, 5.6)

Note:
Probabilities reported as % (95% CI)

Table 4: 10-Year Probability of Stroke for Females

Risk Profile	Age (Years) 40	50	60
Baseline	0.6 (0.1, 1.0)	1.2 (0.6, 1.7)	2.5 (1.4, 3.6)
Diabetes	1.0 (0.0, 2.2)	2.1 (0.0, 4.4)	4.5 (0.0, 9.0)
High BP	1.1 (0.2, 2.0)	2.3 (1.3, 3.4)	4.9 (3.2, 6.6)
High BP + Diabetes	2.0 (0.0, 4.4)	4.2 (0.0, 8.4)	8.7 (0.0, 16.7)
Smoker	0.8 (0.2, 1.4)	1.7 (0.9, 2.5)	3.6 (1.7, 5.4)

Note:
Probabilities reported as % (95% CI)

Table 5: Summary of Blood Pressure and Diabetes Status by Clinical Exam

Characteristic	Sex	Clinical Exam 1	2	3
Number of Observations	Male	1930	1675	1357
	Female	2472	2204	1837
Systolic Blood Pressure (mmHg), mean (SD)	Male	131.6 (19.4)	135.3 (19.9)	138.9 (20.9)
	Female	133.7 (24.3)	137.7 (24.2)	140.6 (24.1)
Diabetes, n (%)	Male	59 (3.1)	74 (4.4)	113 (8.3)
	Female	60 (2.4)	75 (3.4)	130 (7.1)

Discussion & Limitations

- Only smoking status was significantly associated with stroke
  - After adjusting for age, systolic blood pressure and diabetes
- 10-year Probability of Stroke
  - High for men with diabetes
  - Highest for diabetes + high BP for both sexes
- Covariates changed between exams
  - Systolic Blood Pressure increased
  - Diabetes became more prevalent
- Limitations
  - AIC-based backwards selection
  - Static (baseline) covariates used for modeling
    - Future work should consider a time-varying approach